Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/24/2024
PI Name:	Diaz Artiles, Ana Ph.D.		
Project Title:	Predicting Acute Cardiovascular and Ocular Changes due to Changes in the Gravitational Vector and Effects of Countermeasures		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	adartiles@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-909-0644
Organization Name:	Texas A&M University		
PI Address 1:	Aerospace Engineering Department		
PI Address 2:	701 Ross Street		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	09/01/2020	**End Date:**	06/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2025 per V. Lehman/JSC (Ed., 4/20/23) NOTE: End date changed to 8/31/2023 per NSSC information (Ed., 8/22/22) NOTE: End date changed to 8/31/2022 per NSSC information (Ed., 8/13/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dunbar, Bonnie Ph.D. (Texas A&M University)		
Grant/Contract No.:	80NSSC20K1521		
Performance Goal No.:			
Performance Goal Text:			

	Exposure to weightlessness results in the removal of hydrostatic pressure gradients and a permanent headward fluid shift, causing a redistribution of blood. Additionally, in space postural changes don't occur and thus, astronauts are not exposed to daily fluid shifts (between supine and upright postures) as we are on Earth. This has currently unknown consequences, but it might be related to a series of neuro-ocular and functional changes developed in some astronauts during both short and long-duration spaceflight, collectively known as Spaceflight Associated Neuro-Ocular Syndrome (SANS). While the exact etiology of SANS is currently unknown, chronic fluid redistribution affecting intravascular, interstitial, and cerebrospinal fluids and pressures is widely hypothesized to be a contributing factor. Additionally, recently demonstrated stagnant and retrograde blood flow and venous thrombosis in the left internal jugular vein during spaceflight could also be associated with sustained headward blood and tissue fluid shift. Based on the current knowledge and hypotheses, countermeasures focused on producing hydrostatic gradients or reducing the microgravity-induced fluid shift, such as lower body negative pressure (LBNP) or centrifugation, become particularly interesting. LBNP shifts fluids from the head to the lower body is expected to reduce the translaminar pressure across the eye and thus, might prevent some of the ocular changes and remodeling associated with SANS. LBNP is also a promising countermeasure to enhance venous flow in the upper body. Similarly, head-to-foot (Gz) centrifugation also produces a fluid shift and induces hydrostatic gradients within the vessels in the body, which may also reduce intraocular pressure (IOP), reduce the translaminar pressure, and enhance venous flow. However, at present, it is not possible to estimate the overall physiological response (neither acute nor long term) of a particular "dose" of artificial gravity (AG) or LBNP, and more specifically and relevant to our purposes, the response in the upper body and around the eye. As a first step and in the context of this short project, we propose to focus on acute responses. Our objective is to generate acute gravitational dose-response curves of cardiovascular (CV) and ocular variables due to changes in the gravitational vector, with and without countermeasures. We propose to conduct a series of experimental studies (using tilt table, LBNP, and centrifugation), and combine the results with numerical modeling to leverage the advantages of both experimental and computational methodologies. Then, we propose to exert the full flexibility of the numerical framework to investigate CV and ocular responses in additional configurations where data collection is difficult, expensive, or infeasible.
Task Description:	A short-description of the experiments is shown below: - Experiment 1: Tilt table. Subjects will be exposed to multiple tilt angles, from 45° HUT (head up tilt) to 45° HDT (head down tilt), in both prone and supine configurations. CV and ocular measures will be collected and used to generate gravitational dose-response curves as a function of tilt angle. - Experiment 2: LBNP. Subjects will be exposed to multiple levels of negative pressure (from 0 to -50 mmHg), in both supine and 15° HDT. Similarly, CV and ocular measures will be collected and used to generate gravitational dose-response curves as a function of external pressure. - Experiment 3: Centrifugation. Subjects will be exposed to multiple levels of artificial gravity on the human short-radius NASA centrifuge that is being relocated and installed at Texas A&M University (TAMU). CV and ocular measures will be collected and used to generate gravitational dose-response curves as a function of g-level. In addition to the experiments, we will develop a numerical model from two different but complementary lumped-parameter models that have been previously validated and published in the literature: 1) A full-body model that provides an accurate simulation of short-term CV regulatory changes during changes in the gravitational vector (i.e., different tilt angles, centrifugation) and countermeasures (i.e., LBNP, exercise). This model features a more accurate body representation, including a cardiac pacemaker (i.e., pulsatile waveform), baroreceptor and cardiopulmonary feedback control of heart and arterial parameters, and a large number of compartments, particularly in the Gz direction, allowing for an improved representation of hydrostatic gradients in this direction and overall body hemodynamics. 2) An eye model that simulates volume/pressure alterations in the eye during gravitational changes. We will combine both approaches leveraging the advantages of each one of them into a unique more powerful and versatile model to study SANS, the cranial venous system, and other fluid shifts mechanisms and the effects of potential countermeasures. The model will be validated with the experimental data generated during the experiments. Results from this investigation will inform current and future countermeasure development and in-flight prescriptions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project investigates the influence of gravity (or more generally fluid shifts) on cardiovascular and ocular responses using a variety of interventions, including tilt paradigms, Lower Body Negative Pressure (LBNP), and centrifugation. Results will provide critical information for current and future cardiovascular and ocular related countermeasures and in-flight prescriptions. In addition, this research effort has direct application to cardiovascular and ocular pathologies on Earth, for example, orthostatic hypotension and conditions related to cardiovascular regulation.
	The objective of this ground-based research effort is to generate acute gravitational dose-response curves of cardiovascular and ocular variables when exposed to different types of orthostatic stress. We propose to use both experimental and computational approaches to leverage the advantages of each one of these research methodologies. Our study consists of three different ground experiments where the same male and female subjects are exposed to different levels of tilt (from 45° HUT to 45° HDT, experiment 1), different levels of lower body negative pressure or LBNP (from 0 mmHg to -50 mmHg, in both supine and -15° HDT, experiment 2), and centrifugation (from 0g to 2g, measured at the center of mass, experiment 3). In all experiments we are collecting objective measures related to human performance and cardiovascular regulation, including continuous CV variables (Finapres NOVA), non-invasive cardiac output (Innocor, Innovision), ECG and autonomic responses (Finapres NOVA), intraocular pressure (IOP) (I-care tonometer), brachial blood pressure for calibration (Omron), metabolic data (Innocor), blood volume (Blood Tec), Internal Jugular Vein (IJV) pressure (Vein Press), and cross-sectional areas and flows of the IJV (left and right side) and the Common Carotid Arteries (CCA, left and right side) (Ultra sound VScan Extend and Butterfly iQ+). We will derive other metrics such as Ocular Perfusion Pressure. When possible, measurements are continuously monitored during the pre- (baseline), during-, and post-experimental testing (e.g., Finapres); otherwise, measurements are collected after an appropriate time of exposure to the intervention to ensure steady state (typically 5 min). During baseline, all data are also collected in upright seated position. In 2023 we received an additional augmentation grant in order to extend our study from male subjects to also include female subjects (tilt and LBNP) and capture sex-differences in the gravitational dose-response curves.

Task Progress:

To date, we have successfully completed the first two experiments (i.e., tilt and LBNP) for both male and female subjects. During this past year, we have focused on refining our data analysis from experiment 1 and 2, and on preparing the many publications we expect to have from this project. Thus, in 2024 we have already published 2 journal papers related to this project (Hall et al. 2024 and Fois et al 2024), and we have several more in preparation and/or in review. We also continue to work on the Aerospace Engineering centrifuge and expect to have it ready soon, so we can conduct our third experiment.

In addition, in 2024 we received an augmentation grant to investigate cardiovascular and ocular adaptation during 60 minutes of exposure to simulated microgravity (6° HDT). In the last few months, we completed the testing of 24 subjects and data analysis is underway.

In the next few paragraphs, we provide a summary of what has been completed in this project to date.

EXPERIMENTAL WORK

Data from experiment 1 (male subjects) have been captured in the following peer-reviewed publications:

• Whittle R.S. and Diaz-Artiles A. “Gravitational effects on jugular and carotid hemodynamics in graded head up and head down tilt”. Journal of Applied Physiology 2023, 134: 217–229. doi: 10.1152/japplphysiol.00248.2022

• Whittle R.S., Keller N., Hall E.A., Vellore H.S., Stapleton L.M., Findlay K.H., Dunbar B.J., and Diaz-Artiles A. “Gravitational dose response curves for acute cardiovascular hemodynamics in a tilt paradigm”. Journal of the American Heart Association 2022, 11: e024175. doi: 10.1161/JAHA.121.024175 Data analysis of female subjects has also been completed and a journal article is in preparation.

Data analysis from experiment 2 (LBNP) has also been completed and our first peer-reviewed article has been published: • Hall E., Whittle R.S., Diaz-Artiles A. (2024) “Ocular Perfusion Pressure is not Reduced in Response to Lower Body Negative Pressure” npj Microgravity 10, 67. Several journal articles related to our LBNP results are currently in preparation

The complete data are captured in the following dissertation with several journal articles in preparation:

• Whittle R.S., "Quantifying the effects of altered-gravity and spaceflight countermeasures of acute cardiovascular and ocular hemodynamics". Ph.D. Dissertation, Texas A&M University, August 2023.

For experiment 3 (centrifugation), we are planning to use the Aerospace Human Centrifuge at TAMU, which is a NASA-funded facility that was formerly at the University of Texas Medical Branch (UTMB) in Galveston (TX). The final installation and check-out of the centrifuge are currently underway. This process has experienced further delays since year 2, but we expect to have the centrifuge operational soon.

Further, during the course of our data collection for experiments 1 and 2, we published a manuscript covering the differences observed in measurement of cardiac output in altered gravity conditions using inert gas rebreathing and pulse contour analysis. These data are captured in the following peer-reviewed publication:

• Whittle R.S., Stapleton L.M., Petersen L.G., and Diaz-Artiles A. “Indirect measurement of absolute cardiac output during exercise in simulated altered gravity is highly dependent on the method”. Journal of Clinical Monitoring and Computing 2021, 1-12. doi: 10.1007/s10877-021-00769-y

MODELING WORK

We are also investigating gravitational dose responses using modeling approaches. We are developing a comprehensive numerical framework capable of predicting the expected acute cardiovascular responses and ocular changes when exposed to different types of orthostatic stress. We will use our own experimental data to validate the model.

To date, we modified an existing numerical model of the eye (Nelson et al. 2017) to model Intraocular Pressure (IOP) during a full range of tilt positions (360°, which covers both prone and supine positions). Experimental data collected in 13 subjects supported the hypothesis that IOP is statistically significantly higher in prone position than in supine position due to the extra hydrostatic column between the eye globe and the coronal plane, and this is true at most of the tilt angles investigated. Our modified version of the model successfully reproduced these results the results are captured in the following peer-reviewed publication:

• Petersen L.G., Whittle R.S., Lee J.H., Sieker J., Carlson J., Finke C., Shelton C.M., Petersen J.C.G., and Diaz-Artiles A. “Gravitational effects on intraocular pressure and ocular perfusion pressure”. Journal of Applied Physiology 2022, 132: 24-35. doi: 10.1152/japplphysiol.00546.2021

We have extended our full-body model to include a detailed simulation of the head incorporating multiple cranial venous drainage pathways. We have also improved the simulation of the decrease in intravascular volume as a result of transcapillary flow to the dependent vasculature during head-down tilt, as well as the effects of body weight. Validation of these changes has been completed with data from experiments 1 and 2, and the model has also been used to simulate additional scenarios, such as centrifugation, and changes in gravity. Results are being prepared for publication in a peer-reviewed journal and will be shared with the research community.

Additionally, we have conducted a comprehensive sensitive analysis in a “full body” numerical model (Diaz Artiles et al. 2019) to investigate individual differences in cardiovascular responses to orthostatic stress (Whittle and Diaz-Artiles 2021). Conditions simulated include constant gravity conditions (from 0g to 1g, in increments of 0.25g), and gravitation stress generated by a short-radius centrifuge, which induce a strong gravity gradient in the head-to-toe direction (from 0g to 1g, in increments of 0.25g, measured at the center of mass). Results from the constant gravity conditions show that model parameters related to the length, resistance, and compliance of the large veins and parameters related to right ventricular function have the most influence on model outcomes. For most outcome measures considered, parameters related to the heart are dominant. Results highlight which model parameters to accurately value in simulations of individual subjects’ CV response to gravitational stress, improving the accuracy of predictions. Influential parameters remain largely similar across gravity levels, highlighting that accurate model fitting in 1 g can increase the accuracy of predictive responses in reduced gravity. These research efforts have been captured in the following peer-reviewed publication:

• Whittle R.S. and Diaz-Artiles A. “Modeling individual differences in cardiovascular response to gravitational stress

	using a sensitivity analysis". Journal of Applied Physiology 2021, 130: 1983-2001 Similar analysis is currently being conducted in the centrifugation conditions and a manuscript is in preparation. 2024 AUGMENTATION GRANT Finally, as part of the 2024 augmentation grant, we tested 24 subjects to characterize cardiovascular and ocular responses during the first 60 min of 6° HDT. Testing has been finalized and data analysis is underway. In addition, an additional cardiovascular modeling work associated with this research effort was published in npj Microgravity (this work was done in collaboration with our colleagues from the Univeristy of Torino, Italy): • Fois M., Diaz-Artiles A., Zaman S.Y., Ridolfi L., Scarsoglio S. (2024) "Linking cerebral hemodynamics and ocular microgravity-induced alterations through an in-silico/in-vivo head-down tilt framework". npj Microgravity 10, 22.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Diaz Artiles A, Whittle RS. "Cardiovascular dose-response to lower body negative pressure and implications for countermeasure design." Aerospace Medical Association's 94th Annual Scientific Meeting (AsMA), Chicago, Illinois, May 5-9, 2024. Abstract published in Aerosp Med Hum Perform. Vol. 95, No. 8, pp 100, August 2024. , May-2024
Abstracts for Journals and Proceedings	Zaman SY, Fois M, Scarsoglio S, Ridolfi L, Diaz-Artiles A. "The effect of 60-minute head-down tilt on the cross-sectional area of the internal jugular vein." Aerospace Medical Association's 94th Annual Scientific Meeting (AsMA), Chicago, Illinois, May 5-9, 2024. Abstract published in Aerosp Med Hum Perform. Vol. 95, No. 8, pp 46, August 2024. , May-2024
Abstracts for Journals and Proceedings	Whittle RS, Diaz-Artiles A. "Modeling the influence of altered-gravity and orthostatic stress on the cardiovascular system." Aerospace Medical Association's 94th Annual Scientific Meeting (AsMA), Chicago, Illinois, May 5-9, 2024. Abstract published in Aerosp Med Hum Perform. Vol. 95, No. 8, pp 102, August 2024. , May-2024
Articles in Peer-reviewed Journals	Fois M, Diaz-Artiles A, Zaman SY, Ridolfi L, Scarsoglio S. "Linking cerebral hemodynamics and ocular microgravity-induced alterations through an in-silico/in-vivo head-down tilt framework." npj Microgravity. 2024 Feb 27;10:22. https://doi.org/10.1038/s41526-024-00366-8 ; PubMed PMID: 38413627; PubMed Central PMCID: PMC10899661 , Feb-2024
Articles in Peer-reviewed Journals	Hall EA, Whittle RS, Diaz-Artiles A. "Ocular perfusion pressure is not reduced in response to lower body negative pressure." npj Microgravity. 2024 Jun 8;10(1):67. https://doi.org/10.1038/s41526-024-00404-5 ; PubMed PMID: 38851800; PubMed Central PMCID: PMC11162494 , Jun-2024
Conference Materials (Downloadable)	Diaz-Artiles A, Whittle RS, Keller N, Hall EA, Zaman SY, Vellore HS, Patanam S, Dunbar BJ. "Cardiovascular and ocular gravitational dose-response curves and effects of countermeasures." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024 Diaz Artiles_Diaz Artiles_Gravitational Dose Responses.pdf
Conference Materials (Downloadable)	Whittle RS, Diaz-Artiles A. "Computational modeling of cardiovascular dose-response curves in altered-gravity." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024 NASA HRP Abstract – Whittle.pdf
Conference Materials (Downloadable)	Zaman SY, Fois M, Scarsoglio S, Ridolfi L, Diaz-Artiles A. "Cardiovascular and ocular changes observed during 60 minutes of head-down tilt." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024 HRP 2024 Abstract - Zaman.pdf
Dissertations and Theses	Whittle RS. "Quantifying the effects of altered-gravity and spaceflight countermeasures of acute cardiovascular and ocular hemodynamics." PhD Dissertation. Texas A&M University, August 2023. , Aug-2023
Significant Media Coverage	Schaechinger A. (Diaz-Artiles interview). "Researchers learning more about eyes thanks to astronauts. Texas A&M researchers are investigating the impacts of space travel on eyehealth in hopes of counteracting the effects of fluid shifts." Texas A&M Today, June 20, 2024. https://today.tamu.edu/2024/06/20/researchers-learning-more-about-eyes-thanks-to-astronauts/ , Jun-2024